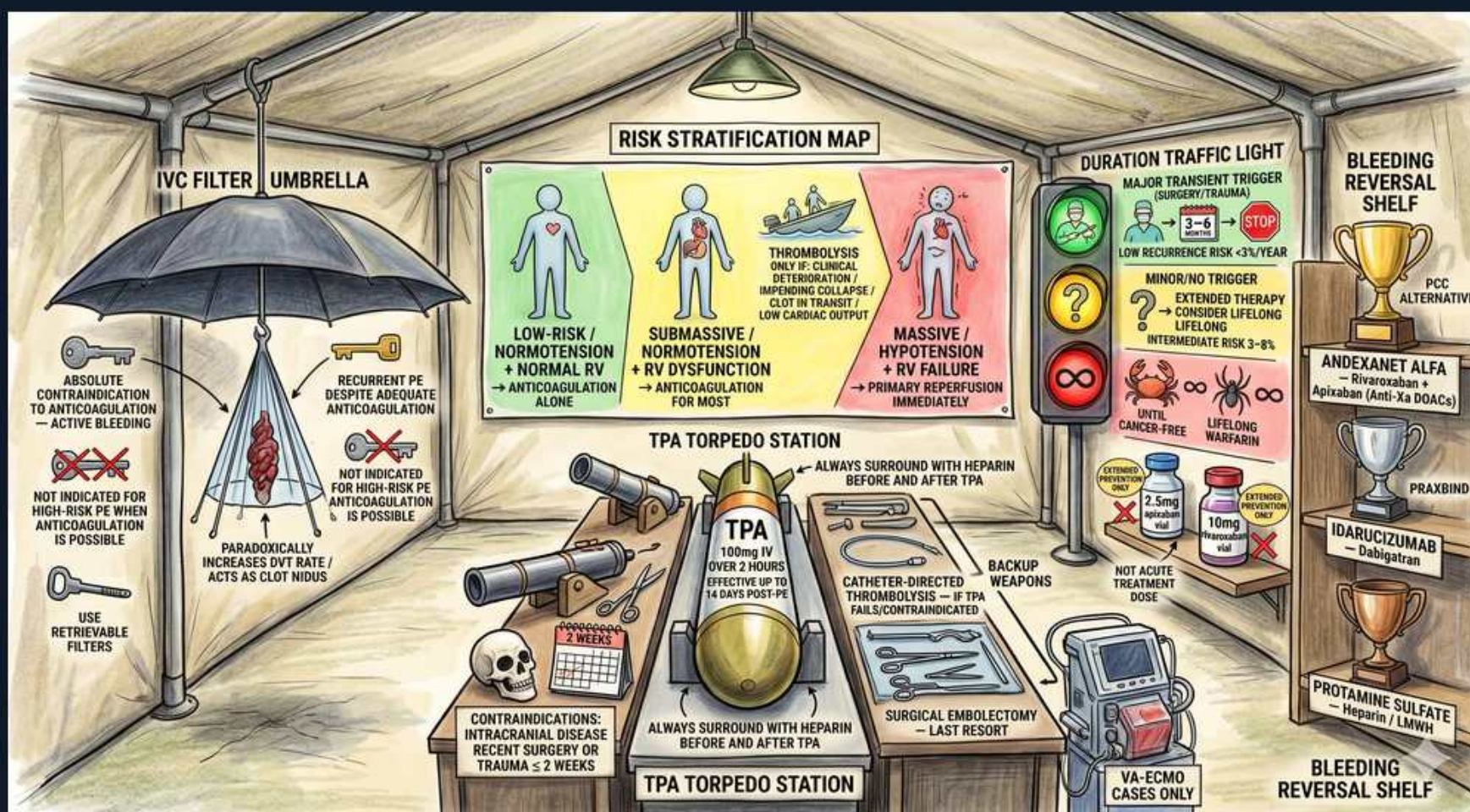


VTE — PE Treatment: tPA, IVC Filter, Risk Tiers & Reversal Agents



1. IVC filter umbrella

WHAT YOU SEE

Umbrella IVC filter

WHAT IT ENCODES

An IVC filter is indicated only when anticoagulation is absolutely contraindicated (active bleeding) or for recurrent PE despite adequate anticoagulation. Retrievable filters are preferred and should be removed once anticoagulation can resume.

BOARD TRAP

IVC filter is NOT a substitute for anticoagulation — it prevents embolization but doesn't treat existing clot; resume anticoagulation and retrieve the filter as soon as safe.

2. Absolute contraindication — active bleeding

WHAT YOU SEE

'Absolute contraindication to anticoagulation — active bleeding'

WHAT IT ENCODES

The core indication for an IVC filter is active bleeding or another absolute contraindication to anticoagulation in a patient with VTE who still needs embolus protection.

BOARD TRAP

Don't place a filter for a routine PE that can be anticoagulated — only when anticoagulation is truly contraindicated.

3. Recurrent PE despite anticoagulation

WHAT YOU SEE

'Recurrent PE despite adequate anticoagulation'

WHAT IT ENCODES

The second filter indication: breakthrough VTE despite therapeutic anticoagulation. First confirm adherence/adequacy before attributing failure.

BOARD TRAP

'Recurrent PE on anticoagulation' → verify therapeutic dosing/compliance first; only then add a filter.

4. Filter paradoxically ↑ DVT

WHAT YOU SEE

'Paradoxically increases DVT rate — acts as clot nidus'

WHAT IT ENCODES

Indwelling filters become a thrombosis nidus, paradoxically increasing the rate of lower-extremity DVT over time — another reason to retrieve early.

BOARD TRAP

Long-term filters RAISE DVT risk — they are not benign; remove retrievable filters once anticoagulation is possible.

5. Risk stratification map

WHAT YOU SEE

Green/yellow/red risk map: low / submassive / massive

WHAT IT ENCODES

Treatment is tier-driven: low-risk (normotensive, normal RV) → anticoagulation alone; submassive (RV dysfunction, normotensive) → anticoagulation, lytics only if deteriorating; massive (hypotension/RV failure) → immediate reperfusion (tPA).

BOARD TRAP

Match therapy to tier: RV strain WITHOUT hypotension is submassive (anticoagulate), only hypotension = massive (thrombolyse).

6. Massive → primary reperfusion

WHAT YOU SEE

Red panel 'massive hypotension + RV failure → primary reperfusion immediately'

WHAT IT ENCODES

Massive (high-risk) PE with hypotension or shock requires immediate systemic thrombolysis (tPA) unless contraindicated; catheter-directed therapy or surgical embolectomy if lytics fail/contraindicated.

BOARD TRAP

Hemodynamic instability is the trigger for thrombolysis — don't wait for arrest; treat the hypotensive PE immediately.

7. tPA — alteplase dose

WHAT YOU SEE

tPA torpedo station, '100mg IV over 2 hr'

WHAT IT ENCODES

Systemic thrombolysis for PE: alteplase 100 mg IV over 2 hours. It rapidly lyses clot and offloads the RV in massive PE. Always overlap/surround with heparin before and after tPA.

BOARD TRAP

Surround tPA with HEPARIN before and after — and tPA is for MASSIVE PE; in stroke the dose/window differ, don't conflate protocols.

8. tPA contraindications

WHAT YOU SEE

Skulls, '≤2 weeks surgery/trauma, intracranial disease'

WHAT IT ENCODES

Absolute thrombolysis contraindications include prior intracranial hemorrhage, recent ischemic stroke, intracranial neoplasm/AVM, active bleeding, and recent surgery/major trauma (≤2-3 weeks).

BOARD TRAP

Recent surgery/trauma (≤2-3 wk) or any intracranial lesion contraindicates systemic lytics → use catheter-directed therapy or surgical embolectomy instead.

9. Catheter-directed thrombolysis

WHAT YOU SEE

'Catheter-directed thrombolysis — tPA via catheter, lower dose'

WHAT IT ENCODES

Catheter-directed thrombolysis delivers low-dose tPA locally, reducing systemic bleeding — useful when systemic lytics are relatively contraindicated or as backup in high-risk PE.

BOARD TRAP

Catheter-directed therapy is the bleeding-sparing 'backup' — choose it over full-dose systemic tPA when bleeding risk is elevated but reperfusion is needed.

10. Surgical embolectomy — last resort

WHAT YOU SEE

'Surgical embolectomy — last resort'

WHAT IT ENCODES

Surgical (or catheter) embolectomy is reserved for massive PE when thrombolysis fails or is contraindicated, often with VA-ECMO support as a bridge.

BOARD TRAP

Embolectomy is the LAST resort — try thrombolysis/catheter-directed first unless absolutely contraindicated or failed.

11. VA-ECMO support

WHAT YOU SEE

'VA-ECMO cases only'

WHAT IT ENCODES

VA-ECMO bridges patients in refractory obstructive shock or arrest from massive PE until clot is cleared by lytics/embolectomy.

BOARD TRAP

ECMO is supportive bridge, not definitive therapy — it buys time for reperfusion, it doesn't dissolve the clot.

12. Duration traffic light

WHAT YOU SEE

Traffic light: provoked vs unprovoked anticoagulation duration

WHAT IT ENCODES

Duration depends on provocation: major transient risk factor → ~3 months; unprovoked or recurrent → extended/indefinite anticoagulation weighing bleeding risk; active cancer → DOAC or LMWH ongoing.

BOARD TRAP

Provoked by a transient factor = 3 months; UNPROVOKED VTE often warrants indefinite anticoagulation — don't auto-stop at 3 months for unprovoked clots.

13. Cancer-associated VTE

WHAT YOU SEE

'Active cancer → warfarin' panel

WHAT IT ENCODES

Cancer-associated VTE is treated with DOACs (apixaban/rivaroxaban/edoxaban) or LMWH for extended duration; LMWH is preferred with GI/GU lesions or drug interactions. Warfarin is less effective here.

BOARD TRAP

For cancer VTE prefer DOAC or LMWH over WARFARIN — warfarin has higher recurrence in malignancy.

14. Andexanet alfa — Xa reversal

WHAT YOU SEE

Bleeding reversal shelf, 'andexanet alfa (anti-Xa DOACs)'

WHAT IT ENCODES

Andexanet alfa reverses factor-Xa-inhibitor DOACs (apixaban, rivaroxaban). Idarucizumab reverses dabigatran (a DTI). PCC is an alternative for Xa-inhibitor bleeding; protamine reverses heparin; vitamin K/PCC reverses warfarin.

BOARD TRAP

Match reversal to drug: andexanet/PCC for Xa-inhibitors, idarucizumab for dabigatran, protamine for heparin — don't cross them.

15. Protamine — heparin reversal

WHAT YOU SEE

'Protamine sulfate — heparin'

WHAT IT ENCODES

Protamine sulfate fully reverses unfractionated heparin and partially reverses LMWH. It does not reverse fondaparinux or DOACs.

BOARD TRAP

Protamine works on HEPARIN/UFH (partial for LMWH) — it does NOT reverse fondaparinux or oral Xa/DTI agents.